

Model Evaluation Report:

Employee Attrition Prediction

1. Objective:

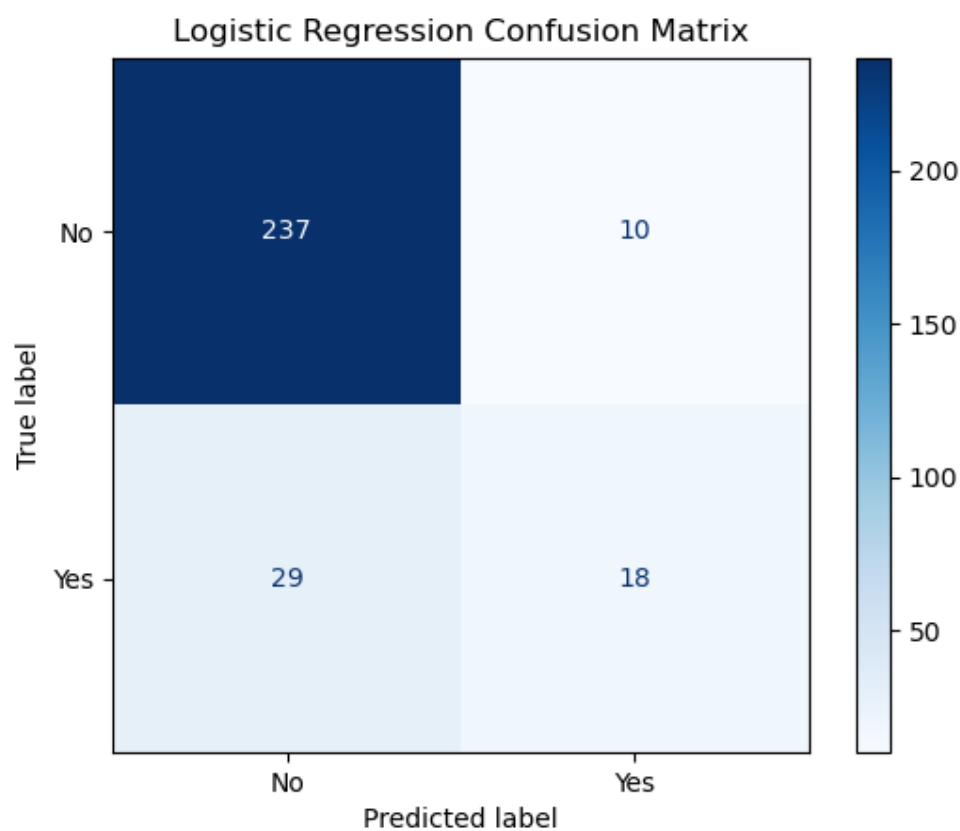
To build machine learning models that predict employee attrition and identify key drivers contributing to it.

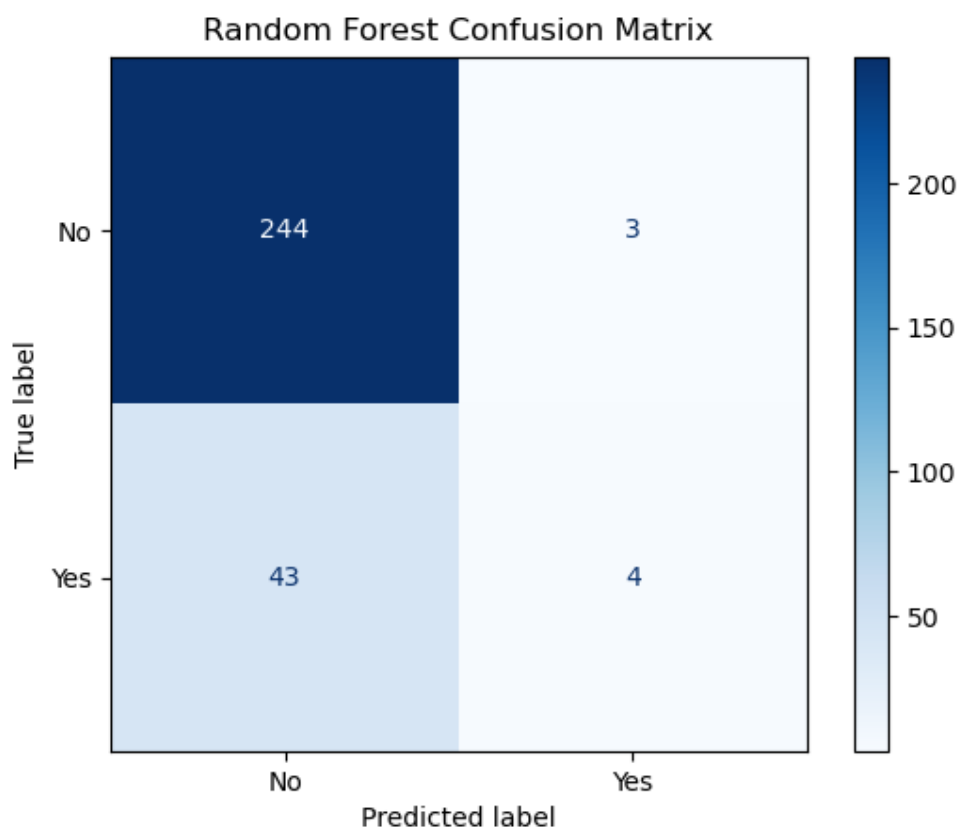
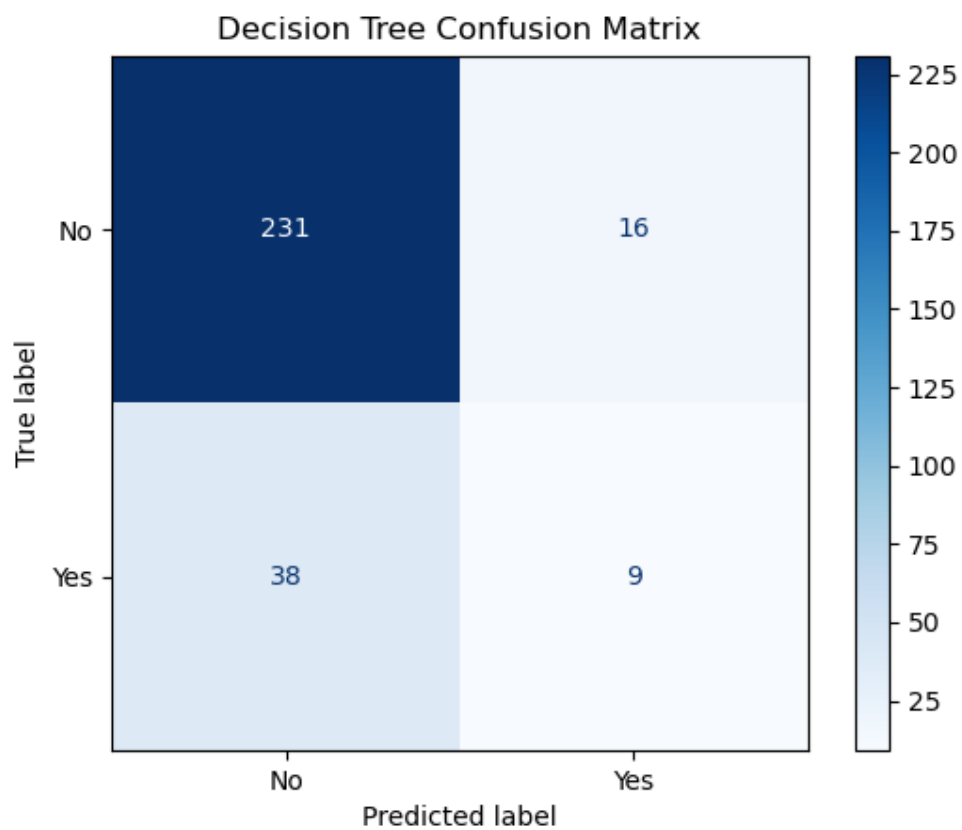
2. Models Compared:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.864	0.845	0.864	0.844
Decision Tree	0.820	0.782	0.820	0.795
Random Forest	0.844	0.806	0.844	0.791

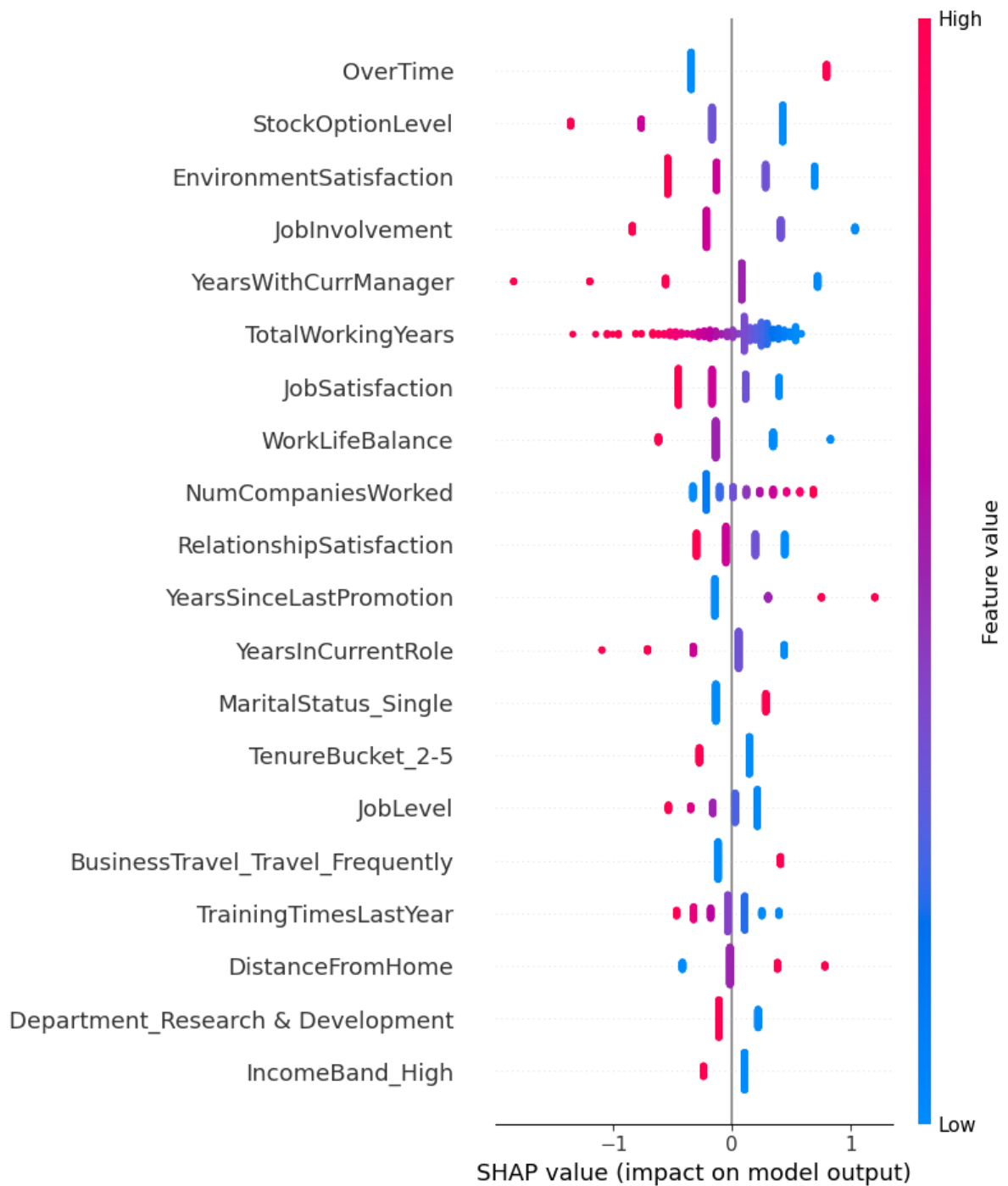
✔ Logistic Regression performed best overall and was selected for further analysis.

3. Confusion Matrices:





4. Feature Importance (SHAP-Based)





Attrition Prevention Suggestions

You can use this writeup directly:

Attrition Prevention Suggestions

1. Key Insights from Data

- Employees working **overtime** have a much higher likelihood of attrition.
- **Low job satisfaction** correlates strongly with employees leaving.
- **Lack of promotions** and career growth are also significant factors.
- Departments like **Sales** and **R&D** show higher attrition trends.

These insights were validated using SHAP explainability and exploratory data analysis.

2. Recommendations

✓ Overtime Monitoring

Implement workload balancing policies to reduce employee burnout. Focus on departments where overtime is most frequent.

✓ Improve Job Satisfaction

Conduct regular surveys to identify dissatisfaction early. Improve recognition programs, work-life balance, and internal communication.

✓ Career Development Programs

Encourage internal promotions, offer training opportunities, and clearly communicate career paths to retain high-potential employees.

✓ Targeted Retention Strategies

Use model predictions to proactively identify high-risk employees and tailor engagement strategies for them.